Do skin releases coincide with favorable champion treatment?

Patch-note attention, balance direction, and player response

The question

League of Legends players sometimes suspect that champions are buffed around a skin release. There are at least three different claims hidden inside that suspicion:

1. Champions receiving a skin may be more likely to appear in the patch notes.
2. When they do appear, their changes may be more likely to be buffs than nerfs.
3. Whatever happens in the patch notes may translate into better in-game performance.

Those claims require different comparisons. This article follows them in order, moving from balance-team attention to the direction of classified changes and finally to observed win, pick, and ban rates.

The result is more nuanced than "skins cause buffs." Champions near a skin release do receive more patch-note attention. However, the evidence does not show a reliable shift toward buffs or an increase in win rate. What changes most clearly after a release is player attention: pick and ban rates rise.

Analysis design

The current analysis uses the following parameters. These are the only non-SQL implementation details shown in the article; the R code that constructs tables, tests, and figures is hidden from the rendered output.

```
skin_release_cutoff_year <- 2013L
skin_patch_window_days <- 42L
champion_release_window_days <- 60L
rework_window_days <- 60L
pre_performance_days <- 28L
post_performance_days <- 14L
```

A skin is eligible when it was released in 2013 or later and is not within 60 days of either the champion's original release or a classified rework. Patch-note windows extend 42 days in both directions from an eligible skin release. Performance changes are measured as the interpolated value 14 days after an anchor minus the value 28 days before it. All rate changes are reported in percentage points.

The unit of the patch-note analysis is a champion–patch opportunity: every champion available on a patch date either appears in that patch's notes or does not. The unit of the balance analysis is one aggregate classification for a champion in a patch. The performance analysis compares eligible skin releases with champion–patch dates outside every eligible skin-release window.

From all skins to the analysis cohort

Table 1: Skin-release cohort construction.

Cohort	Skin releases
All dated skin releases	1,947
Skin releases from 2013	1,510
Skin releases from 2013 outside 60-day champion-release/rework windows	1,420

The cutoff removes the unusually dense early years, while the release/rework exclusion avoids treating the launch of a new champion or a major redesign as an ordinary skin event.

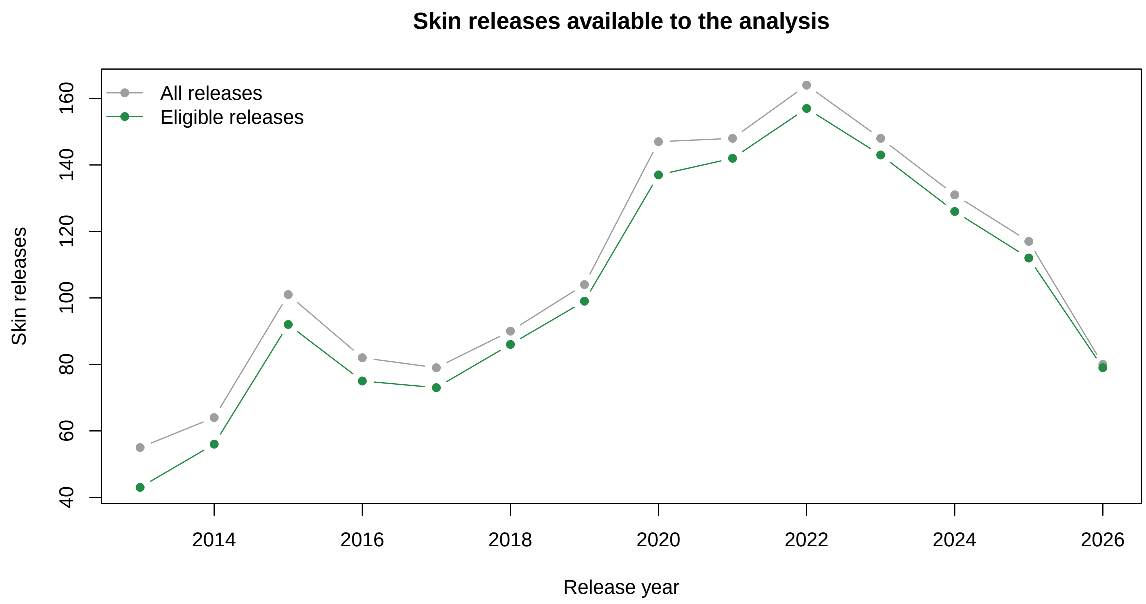

Figure 1: Total and eligible skin releases by year after the analysis cutoff.

Skin releases are not evenly distributed across champions. The champions with the largest catalogues are also a useful warning about confounding: commercially prominent champions recur throughout the data.

Table 2: Champions with the most dated non-base skin releases (top 10 by overall count).

Champion ID	Champion	All releases	Releases since cutoff
21	Miss Fortune	23	17
84	Akali	22	15
99	Lux	22	18
103	Ahri	21	18
81	Ezreal	21	16
22	Ashe	20	15
51	Caitlyn	20	15
1	Annie	19	12
55	Katarina	19	12
64	Lee Sin	19	15

Finding 1: skin releases coincide with more patch-note attention

Champions within a skin-release window appeared in the patch notes more often than other available champions. The overall rates are shown below; the annual plot matters because both the champion roster and Riot's patch philosophy changed over time.

Table 3: Patch-note appearances among champion–patch opportunities.

Skin-release window	Opportunities	Appearances	Appearance rate
No nearby skin release	41,266	7,903	19.2%
Within skin-release window	8,040	1,833	22.8%

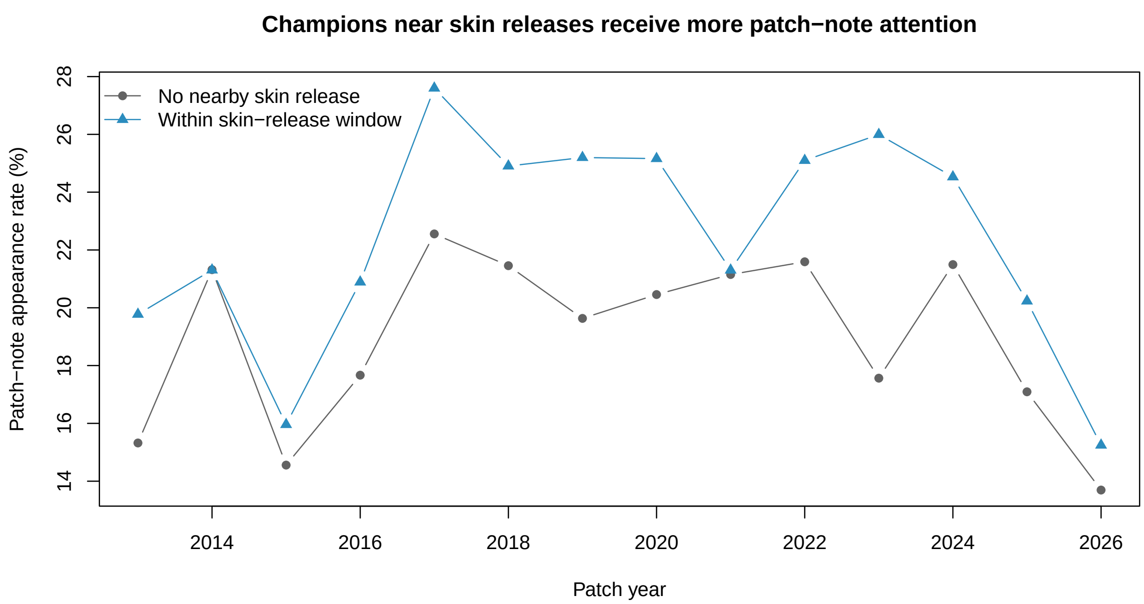

Figure 2: Annual patch-note appearance rates for champions within and outside eligible skin-release windows.

A Mantel–Haenszel test stratified by patch year estimates a common odds ratio of 1.24 (95% CI 1.17–1.32, $p = 2.94\text{e-}13$). In other words, after accounting for year, the odds of appearing in the patch notes are about 24% higher within a skin-release window.

This is strong evidence of an association, but it is not evidence that a skin release causes balance attention. Champions chosen for skins may already be more popular, more actively maintained, or otherwise more likely to need a change.

Finding 2: the extra attention is not clearly more favorable

More patch-note appearances do not necessarily mean more buffs. Among champion–patch classifications labeled as a buff or nerf, the observed buff-to-nerf ratio is somewhat higher near skin releases.

Table 4: Buff and nerf classifications from the cutoff year onward.

Skin-release window	Buffs	Nerfs	Buff-to-nerf ratio	Buff share
No nearby skin release	3,218	2,132	1.51	60.1%
Within skin-release window	672	407	1.65	62.3%

The year-stratified Mantel–Haenszel test estimates a buff odds ratio of 1.08 (95% CI 0.95–1.24, $p = 0.265$). The point estimate leans in the suspected direction, but the confidence interval includes no difference. The data therefore do not provide convincing evidence that the additional attention near skin releases is systematically more buff-oriented.

Finding 3: win rate is stable, while pick and ban rates rise

For every performance observation, the delta is the interpolated rate 14 days after the anchor minus the rate 28 days before it. Positive values represent an increase. The broad comparison population consists of champion–patch dates outside eligible skin-release windows.

Table 5: Mean performance changes from 28 days before to 14 days after the anchor.

Metric	Group	Observations	Mean delta (pp)	Median delta (pp)
Ban rate	General population	38,441	-0.119	-0.038
Ban rate	Skin release	1,366	0.527	0.055
Pick rate	General population	38,441	-0.091	-0.112
Pick rate	Skin release	1,366	1.041	0.556
Win rate	General population	38,441	0.003	-0.017
Win rate	Skin release	1,366	0.022	-0.010

The yearly distributions tell the same broad story. Win-rate changes cluster around zero for both groups. Pick- and ban-rate changes are more dispersed, but skin releases tend to sit higher than the comparison population. Outlying points are suppressed in the figure so that the central distributions remain legible; they remain in every calculation.

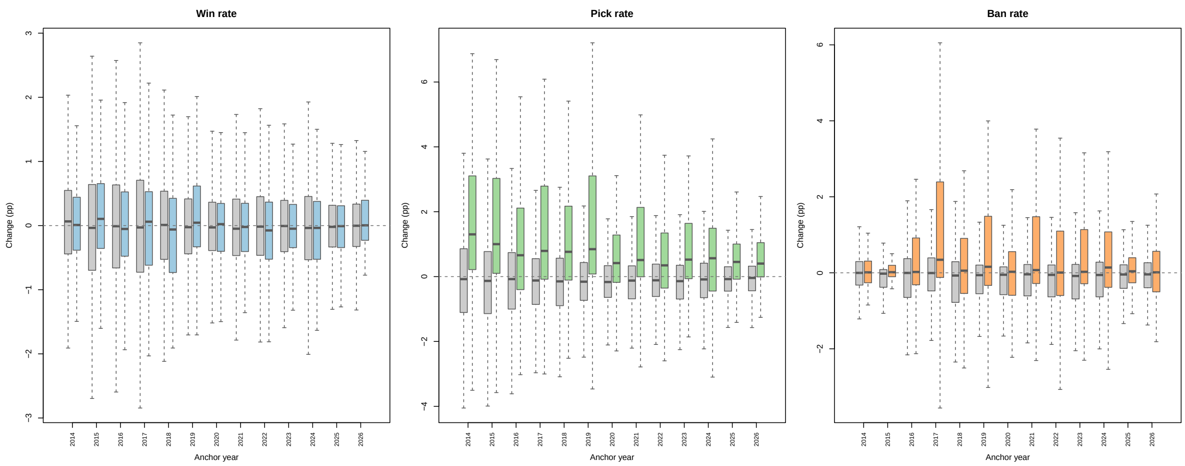

Figure 3: Distributions of after-minus-before changes by anchor year. Grey boxes are the general population; colored boxes are eligible skin releases.

Across all years, eligible skin releases have a mean win-rate delta of +0.022 percentage points, compared with +0.003 in the broad comparison population. The corresponding mean deltas are +1.04 versus -0.09 percentage points for pick rate and +0.53 versus -0.12 for ban rate.

The pick- and ban-rate increases are consistent with players paying more attention to a champion after a cosmetic release. They do not, by themselves, distinguish a skin effect from promotion, concurrent game changes, or selection of champions whose popularity was already moving.

Same-champion controls and popularity

The strongest current win-rate comparison matches every usable skin release to a different patch date for the same champion. A control's complete measurement interval is prohibited from overlapping another skin-release window. This holds fixed the champion without adding hundreds of sparsely observed champion indicator coefficients.

Table 6: Estimated skin-release effect on the win-rate delta under same-champion control models.

Model	Estimated effect (pp)	95% CI	p-value
Same-champion controls, adjusted for year	0.022	-0.055 to 0.098	0.579
Plus pre-window popularity	0.016	-0.060 to 0.092	0.678

The year-adjusted estimate remains close to zero, and adding pre-window popularity does not materially change it. Pre-window popularity itself is negatively associated with the subsequent win-rate delta in this model, but that coefficient is observational and may reflect mean reversion or balance responses to popular champions. It does not support the idea that popularity-driven skin economics produces a detectable short-term win-rate increase.

What the evidence currently says

The clearest result is about **attention**, not favorable treatment. Champions close to a skin release appear in patch notes more often. Once attention is observed, however, the buff/nerf mix is not convincingly different, and neither the broad nor the same-champion comparison detects an average win-rate increase. Pick and ban rates rise after skin releases, suggesting that cosmetic launches coincide with increased player interest even when win rate stays stable.

This distinction matters. "The balance team looks at champions receiving skins" is compatible with the data. "Champions are made stronger to sell skins" is not established by the current results.

What would strengthen the article next

The following additions would contribute more than another descriptive chart:

1. **An event-study figure.** Plot weekly win, pick, and ban-rate differences from several weeks before to several weeks after release, using same-champion placebo dates. This would reveal anticipation, short-lived launch spikes, persistence, and mean reversion that a two-point delta cannot show.
2. **A repeated-measures model for patch-note appearances.** The Mantel–Haenszel test controls for year but treats recurring champion–patch opportunities approximately as independent. A mixed-effects model or clustered standard errors by champion and patch would better reflect the data structure. Pre-window popularity, champion age, and release cadence are natural covariates.
3. **Placebo and permutation checks.** Assign matched pseudo-release dates to the same champions and repeat the complete pipeline. This would show how often effects as large as the observed ones arise from champion selection and calendar structure alone.
4. **Matched analyses for pick and ban rates.** Their descriptive increases are much larger than the win-rate change, but they should receive the same same-champion and popularity-adjusted treatment before being framed as effects.
5. **Skin-level heterogeneity.** Price tier, prestige status, event promotion, simultaneous skin-line launches, and multiple skins for one champion on the same date may separate major commercial releases from routine catalogue additions.

Of these, the event study plus placebo dates would be the most informative next step: it would turn the current before/after snapshot into a time-resolved test while keeping the comparison within champion.

Limitations

- This is an observational analysis. Skin selection, balance attention, champion popularity, and marketing are jointly determined.
- The same champions and patches recur, so simple tests can understate uncertainty unless dependence is modeled.
- Broad performance controls are anchored on patch dates, whereas treatment observations are anchored on exact skin release dates.
- Linear interpolation cannot recover short-lived movements between observations and does not extrapolate beyond a champion series' observed range.
- The classifier reduces all changes for one champion in one patch to a single label. That is the desired unit here, but complex mixed changes inevitably lose detail.
- The latest calendar year may be incomplete and should not be interpreted as a full-year total.

Data queries

The analytical transformations are hidden, but the SQL that defines the source data remains visible for auditability.

Dated non-base skins

```sql
SELECT
  champions.game_id AS champion_id,
  champions.name AS champion_name,
  skins.wiki_skin_id AS skin_id,
  skins.display_name AS skin_name,
  skins.release_date AS skin_release_date,
  base_skin.release_date AS champion_release_date
FROM skins
JOIN champions ON champions.id = skins.champion_id
JOIN skins AS base_skin
  ON base_skin.champion_id = champions.id
 AND base_skin.is_base = 1
WHERE skins.is_base = 0
```

```sql
  AND skins.release_date IS NOT NULL
  AND skins.availability <> 'Canceled'
ORDER BY skins.release_date, champions.name, skins.wiki_skin_id;
```

Performance histories

```sql
SELECT
  champions.game_id AS champion_id,
  graph_series.metric,
  graph_points.observed_at_ms,
  graph_points.value
FROM graph_points
JOIN graph_series ON graph_series.id = graph_points.graph_series_id
JOIN champions ON champions.id = graph_series.champion_id
WHERE graph_series.metric IN ('win_rate', 'popularity', 'ban_rate')
ORDER BY champions.game_id, graph_series.metric, graph_points.observed_at_ms;
```

Aggregate champion–patch classifications

```sql
SELECT
  champions.game_id AS champion_id,
  patch_classifications.patch_id,
  patches.release_date AS patch_release_date,
  patch_classifications.label AS classification_label,
  patch_classifications.confidence
FROM patch_classifications
JOIN champions ON champions.id = patch_classifications.champion_id
JOIN patches ON patches.patch_id = patch_classifications.patch_id
WHERE patch_classifications.label IN ('buff', 'nerf', 'rework');
```

Champion and patch calendars

```sql
SELECT
  champions.game_id AS champion_id,
  champions.name AS champion_name,
  base_skin.release_date AS champion_release_date
```

```sql
FROM champions
JOIN skins AS base_skin
  ON base_skin.champion_id = champions.id
 AND base_skin.is_base = 1
WHERE base_skin.release_date IS NOT NULL;

SELECT patch_id, release_date AS patch_release_date
FROM patches
WHERE release_date IS NOT NULL
ORDER BY release_date, patch_id;
```

Patch-note appearances

```sql
SELECT DISTINCT
  champions.game_id AS champion_id,
  patch_events.patch_id
FROM patch_events
JOIN champions ON champions.id = patch_events.champion_id
JOIN patches ON patches.patch_id = patch_events.patch_id
WHERE patches.release_date IS NOT NULL;
```